Achyutha Kothapalli

Senior AI Engineer

Agentic AI · Python Backend · Data Engineering

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PROFILE

Senior Python engineer with 8+ years of experience building production backend services, data pipelines and **agentic AI tooling** for automotive simulation and autonomous driving (Foretellix, CARIAD/VW, Renault-Nissan). Shipped LLM agents with tool calling, structured outputs and prompt evaluation. Built automation that cut release documentation and rework effort **from 5 developer-weeks to a few hours every release** and simulation triage **from 1 day to 1 hour**. End-to-end owner of release-critical tools, used to working directly with customers, QA and domain experts. Databricks-certified, with hands-on Azure data engineering. Currently leading a 5-person hackathon team.

SKILLS

AI / LLM	Agentic AI, LLM agents & tool calling, MCP (Model Context Protocol), OpenAI API, Claude API, LangChain, prompt engineering & evaluation, structured outputs (Pydantic), guardrails, LLM knowledge bases, Claude Code, Codex
Backend	Python, FastAPI, REST APIs, React, pytest, Docker, Git/GitHub, pre-commit
Data Engineering	PySpark, SQL, Azure Databricks, Azure Data Factory, Azure Synapse Analytics, data modelling, data quality
DevOps & MLOps	Jenkins CI/CD, Terraform (IaC), MLflow
Domain	ADAS & autonomous driving, scenario-based testing, simulation data, MATLAB/Simulink, DSL design (JetBrains MPS)

PROFESSIONAL EXPERIENCE

Senior Software Engineer

06/2022 – 01/2026

Foretellix GmbH · Düsseldorf

- **Built an agentic AI pipeline that generates simulation artefacts from natural-language intent:** LLM agents interpret high-level user requests, then call tools that fill code templates via structured outputs, grounded in a knowledge base of Markdown instruction files.
- **Made LLM output production-reliable** with prompt evaluation, guardrails, schema validation, output checks and hallucination detection, ensuring consistent and reproducible artefacts.
- **Owned a release-critical documentation generation tool end to end**, used by all developers, QA and customers. Automated it in the cloud, **cutting documentation and rework effort from 5 developer-weeks to a few hours every release**.
- Built LLM-powered developer tooling (OpenAI API) and Python backend services for automated KPI-based triage and root-cause analysis of large-scale simulation data, reducing **triage time from 1 day to 1 hour**.
- Designed **FastAPI REST services** that expose KPI, event-level and diagnostic data as structured JSON, so engineering teams and downstream tools trace defects programmatically.
- Automated nightly pipelines on **Jenkins CI/CD and Docker** across dev and QA, and introduced pytest, pre-commit and data quality gates that stop invalid configurations before they reach downstream workflows.
- Drove code reviews and large-scale refactoring. Mentored and trained developers to use the team's tools effectively.

Senior PMT (Process, Methods & Tools) Engineer

09/2020 – 05/2022

CARIAD SE (Volkswagen Group) · Ingolstadt

- **Designed a domain-specific language (DSL) in JetBrains MPS** with supporting Python components that turns complex signal and configuration logic into standardised inputs. The DSL is function-agnostic and usable by any ADAS function.
- Built a Python backend pipeline that converts structured requirements into machine-readable simulation datasets and configuration artefacts, replacing manual preparation.
- Built validated interfaces between generation services and downstream engineering systems, with reusable validation logic that adapts to evolving schemas, parameters and business rules.
- Worked with developers, domain experts and stakeholders to translate functional requirements into maintainable software integrated into automated workflows.

Scenario Test Analyst
04/2020 – 06/2020

Autonomous Intelligent Driving GmbH · Munich

- Specified and executed structured test scenarios for verification and validation of **L4 autonomous driving functions**, evaluating performance against pass/fail criteria for validation sign-off.
- Built and enhanced internal tools for scenario creation and test data management, improving team efficiency within a short ramp-up period.

Senior Engineer
10/2017 – 11/2019

Renault-Nissan Technology & Business Centre India · Chennai

- **Received the CAE Best Performer Award FY18** for improving simulation workflows, methodology and processes.
- Built Python pipelines that automate post-processing, transformation and validation of **ADAS and autonomous driving simulation** outputs across large numbers of runs.
- Replaced manual data preparation with parameterised generation of test and configuration datasets, enabling significantly larger test coverage.
- Introduced standardised data quality checks that catch incomplete or non-compliant results before analysis, and analysed timestamped multi-parameter datasets for root-cause and robustness investigations.

Research Assistant
10/2015 – 01/2017

ICAMS, Ruhr-Universität Bochum · Bochum

- Built a Python pipeline that automatically generates finite element models (RVEs) from statistical microstructure data, replacing manual, error-prone geometry construction.
- Simulated crystal-plasticity response of metal microstructures in ABAQUS, with reproducible Python workflows for validation and analysis.

Earlier Experience

Student Trainee · Huf Hülsbeck & Fürst GmbH & Co. KG, Velbert — 05/2016 – 01/2017

Internship & Master Thesis · HBPO GmbH & Co. KG, Lippstadt — 10/2014 – 08/2015

Graduate Apprentice Trainee · Hindustan Aeronautics Limited, Bangalore — 08/2011 – 08/2012

LEADERSHIP & PROJECTS

Team Lead — Hackathon · Future Careers Festival
10/2026

- **Leading a 5-person team** from idea through architecture to a working prototype, coordinating scope, task split and delivery.
- Stack: FastAPI backend, React frontend, AI-assisted development with Claude Code.

Agentic AI Disengagement Triage · Hackathon Project
2026

- Built an **agentic AI system** that triages autonomous driving disengagements and points engineers to likely root causes.

Azure Databricks Data Platform · Personal Project
2026

- Built an end-to-end data engineering project on Azure Databricks with infrastructure provisioned as code (Terraform).

EDUCATION

M.Sc. Computational Engineering
Ruhr-Universität Bochum · 2012 – 2015

B.Tech Mechanical Engineering
VIT University, Vellore, India · 2007 – 2011

CERTIFICATIONS

Databricks Certified Data Engineer Associate (2026)

Intel Certified Developer – MLOps Professional

iSAQB Certified Professional for Software Architecture – Foundation Level

AWARDS

CAE Best Performer Award FY18 — Renault-Nissan Technology & Business Centre India

LANGUAGES

English: Native

German: Fluent, C1 (B2 certificate)